Bohan Lu

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Education

2023–present **PhD in Electrical and Computer Engineering**, *Duke University*, Durham, NC, expected 2027–2028
Advisor: Kenneth R. Brown

2019–2022 **MSc in Physics**, *Freie Universität Berlin*, Germany
Thesis: *Entanglement dynamics of a many-body localized system*, advisor: Jens Eisert

2015–2019 **BA in Mathematics and Physics**, *Washington University in St. Louis*
Cum Laude, Distinction in Mathematics, High Distinction in Physics. Thesis: *Reduced density operators of finite-sized Heisenberg XXZ chains*

Publications

2026 **Building codes with transversal CCZ using projective geometry and SAT solvers. B. Lu**, K. R. Brown. Preprint, arXiv:2610.09341. Code and data: github.com/jerrylvx/gate2code-public

2024 **Measuring out quasi-local integrals of motion from entanglement. B. Lu**, C. Bertoni, S. J. Thomson, J. Eisert. *Communications Physics* **7**, 17 (2024). doi:10.1038/s42005-023-01478-5, arXiv:2301.01787

Research Experience

2023–present **Doctoral Research**, *Duke University*, Brown group

- Constructed CSS codes with three logical qubits whose logical CCZ gate is implemented by transversal physical $T/T^\dagger$ gates, at thirteen block lengths from $n = 48$ to $n = 496$, by combining projective geometry with SAT solving.
- Proved that no CSS code with three logical qubits, Z-distance at least 3, and a quasi-transversal CCZ gate exists below $n = 39$.
- Released the code, data, and verification programs for every construction (gate2code-public).
- Designed a dynamical error-correcting code from a minimally non-local hypergraph product code (posters at QIP 2025 and QEC 2025).
- Developed stim-floquet, an open-source Python package for Floquet measurement schedules, GF(2) homology checks, and local detector discovery for CSS codes, with Stim circuit output.

Aug 2026–present **Consulting Contractor**, *Xanadu Quantum Technologies*, remote
Nov 2025–Apr 2026 Research on fault-tolerant quantum computing architecture. Supervisor: Eric Sabo.

May–Aug 2026 **Architecture Intern**, *Xanadu Quantum Technologies*, remote
Research on decoding for quantum LDPC codes. Supervisor: Eric Sabo.

Jul–Oct 2022 **Research Intern**, *HQS Quantum Simulations GmbH*, Karlsruhe, Germany
Developed and benchmarked a spectral resolution method based on Hermite polynomials for resonant-level models, documented in an internal report. Supervisor: Peter Schmitteckert.

2021–2022 **Master's Thesis Research**, *Freie Universität Berlin*, Germany

- Proposed a protocol to extract the local integrals of motion of a many-body localized system from its entanglement dynamics, published as the Communications Physics paper above.
- Ran tensor-network simulations on computing clusters and analyzed quantum-information bounds on entanglement growth.

2017–2018 **Greg Delos Fellowship**, *Washington University in St. Louis*
Studied the space of reduced density operators of finite-sized XXZ Heisenberg ground states. Computed their convex set with the Bethe ansatz and DMRG, and searched for phase-transition signatures with a support vector machine. Advisor: Alexander Seidel. Talk at APS March Meeting 2019.

2016–2017 **Undergraduate Research**, *Washington University in St. Louis*
Gauss–Bonnet theorem for two-dimensional closed orbifolds (advisor: Xiang Tang). Maximum-likelihood search for extended halos around TeV blazars (advisors: Manel Errando, Henric Krawczynski).

Talks and Posters

2026 *Building codes with transversal CCZ using projective geometry and SAT solvers.* Talk, Applied Algebra Seminar, Virginia Tech (online), October 2026.

2026 *Boolean functions: applications in quantum error correction and beyond.* Talk, Duke Quantum Center Theory Seminar, September 2026.

2026 *Discovery and classification of small CSS codes supporting quasi-transversal CCZ logical gate.* Poster, QEC 2026.

2025 *A dynamical error-correcting code from a minimally non-local hypergraph product code.* With K. Brown. Posters, QIP 2025 and QEC 2025.

2023 *Improved decoding of quantum LDPC codes using neural-network enhanced belief propagation.* With A. Pesah, J. Roffe, N. Raveendran. Posters, QIP 2023, QEC 2023 (Sydney), and WE-Heraeus Seminar (Bad Honnef).

2022 *Entanglement dynamics in a many-body localized system.* With C. Bertoni, S. Thomson, J. Eisert. Poster, CRC 183 Summer School 2022.

2019 *Reduced density operators of finite-sized Heisenberg XXZ chains.* With A. Seidel. Talk, APS March Meeting 2019.

Skills

AI skills — Agentic orchestration, engineering, and harness building

Scientific computing — Python (NumPy, SciPy, Matplotlib), Stim, Git and GitHub, LaTeX, HPC clusters

Teaching

Spring 2026 Teaching Assistant, Liberata, Duke Bass Connections, Duke University. Mentored student research projects and contributed to research and development on the algorithms team.

Fall 2024 Teaching Assistant, ECE 520 Quantum Engineering, Duke University

Winter 2020 Teaching Assistant, tensor-network seminar, Freie Universität Berlin

2016–2018 Teaching Assistant (Physics 216), grader (Math 4111), and physics help-desk mentor, Washington University in St. Louis

Awards

2026 Duke Graduate School Conference Travel Award